

Table 15. Thermal characteristics

Symbol	Ratings	Value	Unit
T_{STG}	Storage temperature range	-65 to +150	°C
T_J	Maximum junction temperature	150	

5.3 Operating conditions

5.3.1 General operating conditions

Table 16. General operating conditions

Symbol	Parameter	Operating conditions	Min	Typ	Max	Unit
V_{DD}/V_{DDA}	Standard operating voltage	-	2.7 ⁽¹⁾	-	3.6	V
V_{IN}	I/O Input voltage	All I/O except TT_xx	$V_{SS}-0.3$	-	MIN ($V_{DD} + 3.6V$, 5.5 V) ⁽²⁾	
		TT_xx I/O	$V_{SS}-0.3$	-	$V_{DD} + 0.3$	
V_{CAP}	Internal regulator ON	RUN, SLEEP, STOP0 Modes	1.15	1.20	1.26	
		STOP1 Mode	0.9	0.95	1.0	
f_{HCLK}	AHB clock frequency	-	-	-	144	MHz
f_{PCLK}	APB clock frequency	-	-	-	144	
P_D	Power dissipation at $T_A = 85\text{ °C}$ for suffix 6 ⁽³⁾	-	See Section 6.9: Package thermal characteristics for application appropriate thermal resistance and package. Power dissipation is then calculated according ambient temperature (T_A) and maximum junction temperature (T_J) and selected thermal resistance.			mW
	Power dissipation at $T_A = 125\text{ °C}$ for suffix 3 Section 5.3.1: General operating conditions					
T_A	Ambient temperature for suffix 3 version	-	-40	-	125	°C
	Ambient temperature for suffix 6 version	-	-40	-	85	
T_J	Junction temperature range for suffix 3 version	-	-40	-	140	°C
	Junction temperature range for suffix 6 version.	-	-40	-	105	

- When RESET is released, the functionality is guaranteed down to PDR minimum voltage.
- For operation with voltage higher than $V_{DD} + 0.3\text{ V}$, the internal pull-up and pull-down resistors must be disabled. The minimum and maximum input voltage (V_{in}) must comply with the selected peripheral enabled on the given GPIOs. Refer to the respective peripheral characteristics for details.
- If T_A is lower, higher P_D values are allowed as long as T_J does not exceed T_{Jmax} (see [Section 6.9: Package thermal characteristics](#)).